

CSE 435

Software Engineering

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About me

- BS Computer Science, University of Iowa
- MS Computer Science, Tulane University
- PhD Computer Science, Michigan State
 - Pattern Recognition, Artificial Intelligence, Genetic Algorithms

What I've done in industry



Hewlett
Packard

Branch Flow Analyzer

What I've done in industry



Litton Data Systems

Onboard communications system
for the USS Essex, a Wasp class
amphibious assault ship

What I've done in industry



Naval Biodynamics Laboratory

Horizontal accelerator
data acquisition and analysis

What I've done in industry



any AWOS

collect weather data from Automated
Weather Observation Systems
(AWOS) and send to FAA

What I've done in industry



Mackinac
Software

Android Tablet app to identify and
analyze paint and finish defects for
GM vehicles

Goals of the course

- **Introduce Industrial Strength Software Development**
 - formal processes / artifacts for planning, specifying, designing, implementing, and verifying
 - individual and team based development
 - life cycle issues and umbrella activities
- **Introduce Key Foundations Underlying these Activities**
 - requirements engineering
 - software modeling
 - quality assurance

Overview of Course

- ▮ emphasis on analysis and design
- ▮ learn / apply new techniques for software development
- ▮ learn to work with a group
- ▮ improve technical writing skills
- ▮ improve oral presentation skills
- ▮ become up to date on current trends in SE

Structure of the course

- ▯ short homework assignments
- ▯ in class activities and/or “lab” assignments
- ▯ group project
- ▯ two exams - midterm and final
- ▯ oral presentations and prototype demos

How is this course “different”?

- ▮ NOT a programming course
- ▮ Exercises aim to facilitate understanding, solutions, tradeoffs, sensitivity to challenges that affect industrial software development
- ▮ communication, communication, communication
- ▮ written and oral communication skills will be exercised, improved, and assessed
- ▮ teamwork is critical and will be assessed

Relationship to other courses

Not a design / programming course (contrast with CSE 335)

- higher level coverage of notation
- emphasis is on process rather than design methods
- about much more than producing code

Not a capstone design experience (contrast with CSE 498)

- smaller more constrained project (I think!)
- smaller teams (?)
- end result not necessarily a working product

A few words about flexibility

Now stand up and stretch

```
int op_time = (day == WE) ? 12 : 9;
```

```
int op_time;  
if (day == WE)  
    op_time = 12;  
else  
    op_time = 9;
```

In Industry,

You are writing code
FOR OTHER PEOPLE TO READ

For next Wednesday

- ▯ answers to Background Survey (typed)
- ▯ be prepared to share and discuss some of your answers in class
- ▯ read Chapter 1 in Pfleeger